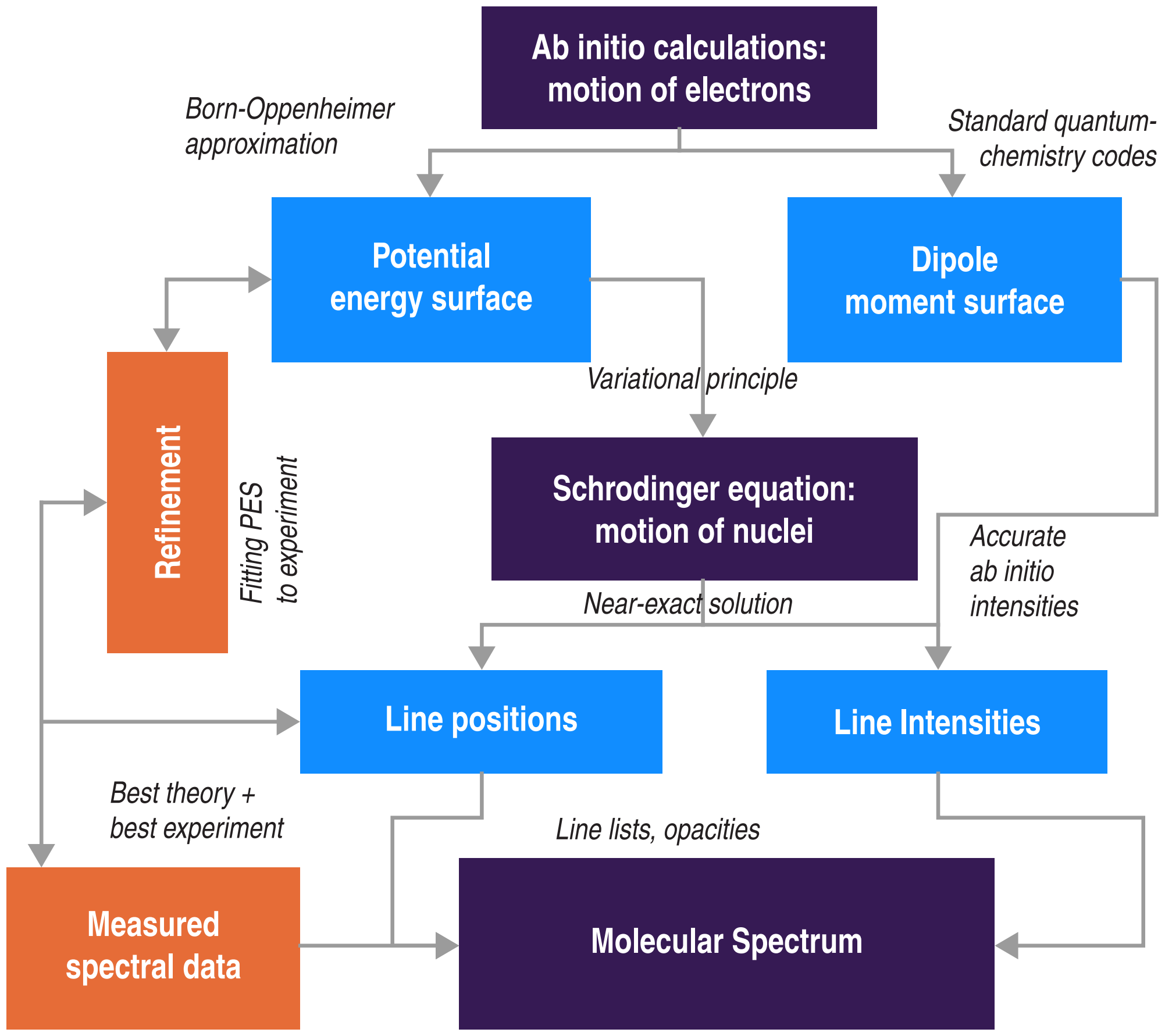
Ab initio calculations:
motion of electrons
Born-Oppenheimer
approximation
Standard quantum-
chemistry codes
Potential
energy surface
Dipole
moment surface
Variational principle
Refinement
Fitting PES
to experiment
Schrodinger equation:
motion of nuclei
Accurate
ab initio
intensities
Near-exact solution
Line positions
Line Intensities
Best theory +
best experiment
Line lists, opacities
Measured
spectral data
Molecular Spectrum